



Lab Report 7
CSE3023 - Web Application and Development

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Task 1

StudentBean.java

```
1
2 package com.lab.bean;
3
4 /**
5  *
6  * @author Nazmi
7  */
8 public class StudentBean implements java.io.Serializable {
9
10     private String matricNo;
11     private String fullname;
12     private String password;
13     private String base64Image;
14
15     public StudentBean() {
16     }
17
18     public String getMatricNo() {
19         return matricNo;
20     }
21
22     public void setMatricNo(String matricNo) {
23         this.matricNo = matricNo;
24     }
25
26     public String getFullname() {
27         return fullname;
```

```
30     public void setFullname(String fullname) {
31         this.fullname = fullname;
32     }
33
34     public String getPassword() {
35         return password;
36     }
37
38     public void setPassword(String password) {
39         this.password = password;
40     }
41
42     public String getBase64Image() {
43         return base64Image;
44     }
45
46     public void setBase64Image(String base64Image) {
47         this.base64Image = base64Image;
48     }
49 }
```

StudentDao.java

```
1  package com.lab.dao;
2
3  /**
4   *
5   * @author Nazmi
6   */
7  import com.lab.bean.StudentBean;
8  import java.io.ByteArrayOutputStream;
9  import java.io.InputStream;
10 import java.sql.Blob;
11 import java.sql.Connection;
12 import java.sql.DriverManager;
13 import java.sql.PreparedStatement;
14 import java.sql.ResultSet;
15 import java.util.Base64;
16
17 public class StudentDAO {
18
19     private Connection getConnection() throws Exception {
20         Class.forName("com.mysql.cj.jdbc.Driver");
21         return DriverManager.getConnection("jdbc:mysql://localhost:3306/lab7_db",
22             "root", "admin");
23     }
24
25     // --- REGISTER LOGIC ---
26     public boolean registerStudent(StudentBean student, InputStream imageStream) {
27         try (Connection conn = getConnection()) {
28             String sql
29                 = "INSERT INTO students (matric_no, password, fullname, profile_image) VALUES (?, ?, ?, ?)";
30             PreparedStatement pstmt = conn.prepareStatement(sql);
31             pstmt.setString(1, student.getMatricNo());
32             pstmt.setString(2, student.getPassword());
33             pstmt.setString(3, student.getFullname());
34             pstmt.setBlob(4, imageStream); // Set BLOB data
35             return pstmt.executeUpdate() > 0;
36         } catch (Exception e) {
37             e.printStackTrace();
38             return false;
39         }
40     }
41     // --- LOGIN LOGIC ---
42
43     public StudentBean loginStudent(String matricNo, String password) {
44         StudentBean student = null;
45         try (Connection conn = getConnection()) {
46             String sql = "SELECT * FROM students WHERE matric_no = ? AND password = ?";
47             PreparedStatement pstmt = conn.prepareStatement(sql);
48             pstmt.setString(1, matricNo);
49             pstmt.setString(2, password);
50
51             ResultSet rs = pstmt.executeQuery();
52             if (rs.next()) {
53                 student = new StudentBean();
54                 student.setMatricNo(rs.getString("matric_no"));
55                 student.setFullname(rs.getString("fullname"));
56                 // Fetch BLOB and convert to Base64 String
57                 Blob blob = rs.getBlob("profile_image");
58                 if (blob != null) {
59                     InputStream inputStream = blob.getBinaryStream();
60                     ByteArrayOutputStream outputStream = new ByteArrayOutputStream();
61                     byte[] buffer = new byte[4096];
62                     int bytesRead = -1;
63                     while ((bytesRead = inputStream.read(buffer)) != -1) {
64                         outputStream.write(buffer, 0, bytesRead);
65                     }
66                     byte[] imageBytes = outputStream.toByteArray();
67                     student.setBase64Image(Base64.getEncoder().encodeToString(imageBytes));
68                 }
69             } catch (Exception e) {
70                 e.printStackTrace();
71             }
72             return student;
73         }
```

```

74 // --- DELETE LOGIC ---
75
76 public boolean deleteStudent(String matricNo) {
77     try (Connection conn = getConnection()) {
78         String sql = "DELETE FROM students WHERE matric_no = ?";
79         PreparedStatement pstmt = conn.prepareStatement(sql);
80         pstmt.setString(1, matricNo);
81         return pstmt.executeUpdate() > 0;
82     } catch (Exception e) {
83         e.printStackTrace();
84         return false;
85     }
86 }
87 };

```

Task 2(User Servlet)

```

1 package com.lab.controller;
2
3 import com.lab.bean.StudentBean;
4 import com.lab.dao.StudentDAO;
5 import java.io.IOException;
6 import java.io.InputStream;
7 import javax.servlet.ServletException;
8 import javax.servlet.annotation.MultipartConfig;
9 import javax.servlet.http.HttpServlet;
10 import javax.servlet.http.HttpServletRequest;
11 import javax.servlet.http.HttpServletResponse;
12 import javax.servlet.http.HttpSession;
13 import javax.servlet.http.Part;
14
15 @MultipartConfig(maxFileSize = 16177215) // max 16 MB
16 public class UserServlet extends HttpServlet {
17     private StudentDAO studentDAO = new StudentDAO();
18
19     @Override
20     protected void doPost(HttpServletRequest request, HttpServletResponse response)
21         throws ServletException, IOException {
22
23         response.setContentType("text/html;charset=UTF-8");
24

```

```

26 // Read hidden parameter
27 String action = request.getParameter("action");
28
29 if ("register".equals(action)) {
30
31     // HANDLE REGISTRATION
32     StudentBean newStudent = new StudentBean();
33
34     newStudent.setMatricNo(request.getParameter("matricNo"));
35     newStudent.setFullname(request.getParameter("fullname"));
36     newStudent.setPassword(request.getParameter("password"));
37
38     InputStream inputStream = null;
39
40     Part filePart = request.getPart("profileImage");
41
42     if (filePart != null && filePart.getSize() > 0) {
43         inputStream = filePart.getInputStream();
44     }
45
46     boolean isSuccess =
47         studentDAO.registerStudent(newStudent, inputStream);
48
49     if (isSuccess) {
50
51         response.getWriter().println(
52             "Registration Successful! "
53             + "<a href='login.html'>Login here</a>"
54         );

```

```

56     } else {
57         response.getWriter().println("Registration Failed!");
58     }
59
60 } else if ("login".equals(action)) {
61
62     // HANDLE LOGIN
63     String matric = request.getParameter("matricNo");
64     String pass = request.getParameter("password");
65
66     StudentBean student =
67         studentDAO.loginStudent(matric, pass);
68
69     if (student != null) {
70
71         // Login successful
72         HttpSession session = request.getSession();
73
74         session.setAttribute("loggedUser", student);
75
76         response.sendRedirect("dashboard.jsp");
77
78     } else {
79
80         response.getWriter().println(
81             "Invalid Credentials! "
82             + "<a href='login.html'>Try Again</a>"
83         );
84
85     }
86
87 }

```

```

91      @Override
92      protected void doGet(HttpServletRequest request,
93                          HttpServletResponse response)
94                          throws ServletException, IOException {
95
96          String action = request.getParameter("action");
97
98          HttpSession session = request.getSession(false);
99
100         if ("logout".equals(action)) {
101
102             // HANDLE LOGOUT
103             if (session != null) {
104                 session.invalidate();
105             }
106
107             response.sendRedirect("login.html");
108
109         } else if ("delete".equals(action)) {
110
111             // HANDLE DELETE ACCOUNT
112             if (session != null
113                 && session.getAttribute("loggedUser") != null) {
114
115                 StudentBean user =
116                     (StudentBean) session.getAttribute("loggedUser");
117
118                 studentDAO.deleteStudent(user.getMatricNo());
119
120                 session.invalidate();
121             }
122
123             response.sendRedirect("register.html");
124         }
125     }
126 }

```

Task 3

Register.htm;

```

1  <!DOCTYPE html>
2  <!--
3  Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
4  Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this template
5  -->
6  <!DOCTYPE html>
7  <html>
8      <head><title>Student Registration</title></head>
9      <body>
10         <h2>Register New Student</h2>
11         <form action="UserServlet" method="POST" enctype="multipart/form-data">
12             <input type="hidden" name="action" value="register">
13             Matric No: <input type="text" name="matricNo" required><br><br>
14             Full Name: <input type="text" name="fullname" required><br><br>
15             Password: <input type="password" name="password" required><br><br>
16             Profile Picture: <input type="file" name="profileImage" accept="image/*" required>
17             <br>
18             <br>
19             <input type="submit" value="Register">
20         </form>
21     </body>
22 </html>
23

```

Login.html

```
1  <!DOCTYPE html>
2  <html>
3  <body>
4      <h2>Student Login</h2>
5      <form action="UserServlet" method="POST">
6          <input type="hidden" name="action" value="login">
7          Matric No: <input type="text" name="matricNo" required><br><br>
8          Password: <input type="password" name="password" required><br><br>
9          <input type="submit" value="Login">
10     </form>
11 </body>
</html>
```

Dashboard.jsp

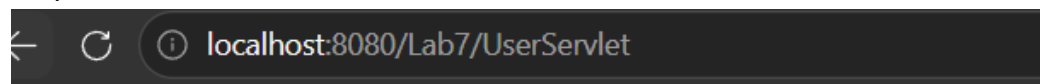
```
1  <!--
2      Document      : dashboard
3      Created on   : 19 May 2026, 3:49:31 pm
4      Author      : Nazmi
5  -->
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <%@page import="com.lab.bean.StudentBean"%>
9  <!DOCTYPE html>
10 <html lang="en">
11 <head>
12     <meta charset="UTF-8">
13     <meta name="viewport" content="width=device-width, initial-scale=1">
14     <title>Student Dashboard</title>
15     <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
16           rel="stylesheet">
17     <style>
18         /* Custom CSS minor untuk saiz dan bingkai gambar sahaja */
19         .profile-img {
20             width: 160px;
21             height: 160px;
22             object-fit: cover;
23             border: 5px solid #198754; /* Warna hijau success Bootstrap */
24         }
25     </style>
26 </head>
27 <body class="bg-light">
```

```

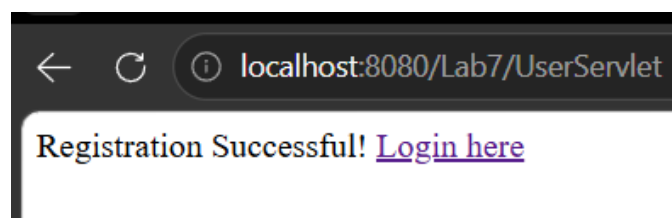
28 <%
29 // 1. Session Verification (Security Check)
30 response.setContentType("text/html;charset=UTF-8");
31 StudentBean user = (StudentBean) session.getAttribute("loggedUser");
32 if (user == null) {
33     response.sendRedirect("login.html");
34     return;
35 }
36 %>
37 <div class="container d-flex justify-content-center align-items-center"
38     style="min-height: 100vh;">
39     <div class="card shadow-lg border-0 rounded-4 p-4 p-md-5 text-center"
40         style="max-width: 600px; width: 100%;">
41         <div class="card-body">
42             <h2 class="fw-bold text-dark mb-2">Welcome, <%= user.getFullname() %>!</h2>
44             <p class="text-muted fs-5 mb-4">Matric Number: <strong class="text-secondary"><div class="d-flex justify-content-center gap-3 flex-wrap">
45                 <a href="UserServlet?action=logout" class="btn btn-dark px-4 py-2
46                     fw-semibold">Logout</a>
47                 <a href="UserServlet?action=delete"
48                     onclick="return confirm('WARNING: Are you sure you want to delete
49                     your account permanently?');"
50                     class="btn btn-danger px-4 py-2 fw-semibold">Delete Account</a>
51             </div>
52         </div>
53     </div>
54 </div>
55 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/
56     bootstrap.bundle.min.js"></script>
57 </body>

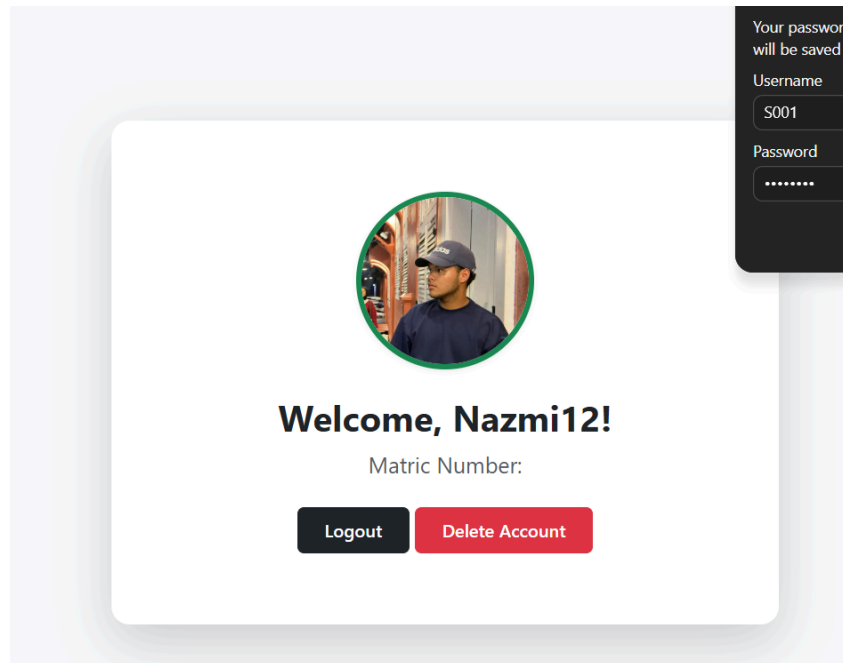
```

Output:



Registration Failed!





Lab Exercise

Registered_subject

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	int(11)			No	None		AUTO_INCREMENT	Change
☐	2	matric_no	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change
☐	3	subject_code	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change
☐	4	subject_name	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change

Student

					matric_no	password
☐		Edit		Copy		Delete
					S001	123

Source Code:

1.registerSubject.jsp

```
6      <!DOCTYPE html>
7      <html>
8      <head>
9          <title>Register Subject</title>
10     </head>
11     <body>
12         <h2>Register Subject</h2>
13         <form action="<%=request.getContextPath() %>/SubjectServlet?action=add"
14             method="post">
15
16             Subject Code:
17             <input type="text" name="subjectCode" required>
18             <br><br>
19             Subject Name:
20             <input type="text" name="subjectName" required>
21
22             <br><br>
23             <input type="submit" value="Register">
24
25         </form>
26
27         <br>
28         <a href="<%=request.getContextPath() %>/SubjectServlet?action=view">
29             View Subjects
30         </a>
31
32     </body>
33 </html>
```

2.updateSubject.jsp

```
1      <%@page import="com.lab.bean.SubjectBean"%>
2
3      <%
4          SubjectBean sb
5              = (SubjectBean) request.getAttribute("subject");
6      %>
7
8      <!DOCTYPE html>
9      <html>
10     <head>
11         <title>Update Subject</title>
12     </head>
13
14     <body>
15
16         <h2>Update Subject</h2>
17
18         <form action="<%=request.getContextPath() %>/SubjectServlet?action=update"
19             method="post">
20
21             <input type="hidden"
22                 name="id"
23                 value="<%=sb.getId() %>">
24
25             Subject Code:
26
27             <input type="text"
28                 name="subjectCode"
29                 value="<%=sb.getSubjectCode() %>">
30
31             <br><br>
32
33             Subject Name:
34
35             <input type="text"
36                 name="subjectName"
37                 value="<%=sb.getSubjectName() %>">
38
39             <br><br>
40
41             <input type="submit"
42                 value="Update">
43
44         </form>
45
46     </body>
47 </html>
```

3. ViewSubject.jsp

```
1 <%@page import="java.util.ArrayList"%>
2 <%@page import="com.lab.bean.SubjectBean"%>
3
4 <%
5 ArrayList<SubjectBean> list =
6     (ArrayList<SubjectBean>)
7     request.getAttribute("subjectList");
8 %>
9
10 <!DOCTYPE html>
11 <html>
12 <head>
13 <title>View Subjects</title>
14 </head>
15
16 <body>
17
18 <h2>Subject List</h2>
19
20 <a href="<%=request.getContextPath() %>/Subject/registerSubject.jsp"
21 Add Subject
22 </a>
23
24 <br><br>
25
26 <table border="1">
```

```
27 <tr>
28 <th>ID</th>
29 <th>Code</th>
30 <th>Name</th>
31 <th>Action</th>
32 </tr>
33
34 <%
35 for (SubjectBean sb : list) {
36 %>
37 <tr>
38
39 <td><%=sb.getId() %></td>
40 <td><%=sb.getSubjectCode() %></td>
41 <td><%=sb.getSubjectName() %></td>
42
43 <td>
44 <a href="<%=request.getContextPath() %>/SubjectServlet?action=edit&id=<%=sb.getId() %>"
45 Edit
46 </a>
47
48 <a href="<%=request.getContextPath() %>/SubjectServlet?action=delete&id=<%=sb.getId() %>"
49 onclick="return confirm('Delete this subject?');">
50 Delete
51 </a>
52 </td>
53 </tr>
54 }
```

```
27 <tr>
28 <th>ID</th>
29 <th>Code</th>
30 <th>Name</th>
31 <th>Action</th>
32 </tr>
33
34 <%
35 for (SubjectBean sb : list) {
36 %>
37 <tr>
38
39 <td><%=sb.getId() %></td>
40 <td><%=sb.getSubjectCode() %></td>
41 <td><%=sb.getSubjectName() %></td>
42
43 <td>
44 <a href="<%=request.getContextPath() %>/SubjectServlet?action=edit&id=<%=sb.getId() %>"
45 Edit
46 </a>
47
48 <a href="<%=request.getContextPath() %>/SubjectServlet?action=delete&id=<%=sb.getId() %>"
49 onclick="return confirm('Delete this subject?');">
50 Delete
51 </a>
52 </td>
53 </tr>
54 }
```

4. Dashboard.jsp

```
1 <%@page contentType="text/html" pageEncoding="UTF-8"%>
2 <%
3 String matricNo
4     = (String) session.getAttribute("matricNo");
5
6 if (matricNo == null) {
7     response.sendRedirect(request.getContextPath()+"/login.html");
8 }
9 %>
10 <!DOCTYPE html>
11 <html>
12 <head>
13 <title>Dashboard</title>
14 </head>
15
16 <body>
17
18 <h1>Student Dashboard</h1>
19 <hr>
20 <h3>
21 Welcome <%= matricNo %>
22 </h3>
23 <hr>
24
25 <a href="Subject/registerSubject.jsp">
```

```
26 Register Subject
27 </a>
28 <br><br>
29 <a href="<%=request.getContextPath() %>/SubjectServlet?action=view">
30 View Subjects
31 </a>
32 </body>
33 </html>
```

5.Login.html

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Login</title>
5  </head>
6  <body>
7      <h2>Student Login</h2>
8      <form action="LoginServlet" method="post">
9
10         Matric No:
11
12         <input type="text"
13             name="matricNo">
14
15         <br><br>
16         Password:
17         <input type="password"
18             name="password">
19
20         <br><br>
21         <input type="submit"
22             value="Login">
23     </form>
24
25 </body>
</html>
```

6. SubjectBean.java

```
1  package com.lab.bean;
2
3  public class SubjectBean {
4
5      private int id;
6      private String matricNo;
7      private String subjectCode;
8      private String subjectName;
9
10     public SubjectBean() {
11     }
12
13     public int getId() {
14         return id;
15     }
16
17     public void setId(int id) {
18         this.id = id;
19     }
20
21     public String getMatricNo() {
22         return matricNo;
23     }
24
25     public void setMatricNo(String matricNo) {
26         this.matricNo = matricNo;
27     }
28 }
```

```

29     public String getSubjectCode() {
30         return subjectCode;
31     }
32
33     public void setSubjectCode(String subjectCode) {
34         this.subjectCode = subjectCode;
35     }
36
37     public String getSubjectName() {
38         return subjectName;
39     }
40
41     public void setSubjectName(String subjectName) {
42         this.subjectName = subjectName;
43     }
44 }

```

7.loginServlet

```

1  package com.lab.controller;
2
3  import java.io.IOException;
4  import java.sql.*;
5  import javax.servlet.*;
6  import javax.servlet.http.*;
7
8  public class LoginServlet extends HttpServlet {
9
10     protected void processRequest(HttpServletRequest request,
11                                   HttpServletResponse response)
12         throws ServletException, IOException {
13
14         String matricNo =
15             request.getParameter("matricNo");
16
17         String password =
18             request.getParameter("password");
19
20         try {
21
22             Class.forName("com.mysql.jdbc.Driver");
23
24             Connection con =
25                 DriverManager.getConnection(
26                     "jdbc:mysql://localhost:3306/lab_7exercise",
27                     "root",
28                     "admin"
29                 );
30
31         } catch (Exception e) {
32             e.printStackTrace();
33         }
34     }
35 }

```

```

31         String sql =
32             "SELECT * FROM students "
33             + "WHERE matric_no=? AND password=?";
34
35         PreparedStatement ps =
36             con.prepareStatement(sql);
37
38         ps.setString(1, matricNo);
39         ps.setString(2, password);
40
41         ResultSet rs =
42             ps.executeQuery();
43
44         if (rs.next()) {
45
46             HttpSession session =
47                 request.getSession();
48
49             session.setAttribute(
50                 "matricNo",
51                 matricNo
52             );
53
54             response.sendRedirect(
55                 "dashboard.jsp"
56             );

```

```

58         } else {
59
60             response.getWriter().println(
61                 "Login Failed"
62             );
63         }
64
65         con.close();
66
67         } catch (Exception e) {
68             e.printStackTrace();
69         }
70     }
71
72     @Override
73     protected void doGet(HttpServletRequest request,
74                           HttpServletResponse response)
75         throws ServletException, IOException {
76
77         processRequest(request, response);
78     }
79
80     @Override
81     protected void doPost(HttpServletRequest request,
82                           HttpServletResponse response)
83         throws ServletException, IOException {
84
85         processRequest(request, response);
86     }
87 }

```

8.SubjectServlet.java

```
package com.lab.controller;

import com.lab.bean.SubjectBean;
import com.lab.dao.SubjectDAO;
import java.io.IOException;
import java.util.ArrayList;
import javax.servlet.*;
import javax.servlet.http.*;

public class SubjectServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        String action = request.getParameter("action");

        HttpSession session = request.getSession(false);

        if (session == null) {
            response.sendRedirect(request.getContextPath() + "/login.html");
            return;
        }

        String matricNo =
            (String) session.getAttribute("matricNo");

        if (matricNo == null) {
            response.sendRedirect(request.getContextPath() + "/login.html");
            return;
        }

        SubjectDAO dao = new SubjectDAO();

        // ADD
        if ("add".equals(action)) {

            SubjectBean sb = new SubjectBean();

            sb.setMatricNo(matricNo);
            sb.setSubjectCode(request.getParameter("subjectCode"));
            sb.setSubjectName(request.getParameter("subjectName"));

            dao.addSubject(sb);

            response.sendRedirect(
                request.getContextPath()
                + "/SubjectServlet?action=view");
        }
    }
}
```

```

// VIEW
else if ("view".equals(action)) {

    ArrayList<SubjectBean> list =
        dao.getSubjects(matricNo);

    request.setAttribute("subjectList", list);

    RequestDispatcher rd =
        request.getRequestDispatcher(
            "/Subject/viewSubjects.jsp");

    rd.forward(request, response);
}

// EDIT
else if ("edit".equals(action)) {

    int id =
        Integer.parseInt(
            request.getParameter("id"));

    SubjectBean sb =
        dao.getSubjectById(id);

    request.setAttribute("subject", sb);

    RequestDispatcher rd =
        request.getRequestDispatcher(
            "/Subject/updateSubject.jsp");

    rd.forward(request, response);
}

// UPDATE
else if ("update".equals(action)) {

    SubjectBean sb =
        new SubjectBean();

    sb.setId(
        Integer.parseInt(
            request.getParameter("id")));

    sb.setSubjectCode(
        request.getParameter("subjectCode"));

    sb.setSubjectName(
        request.getParameter("subjectName"));
}

```



```

        dao.updateSubject(sb);

        response.sendRedirect(
            request.getContextPath()
            + "/SubjectServlet?action=view");
    }

    // DELETE
    else if ("delete".equals(action)) {

        int id =
            Integer.parseInt(
                request.getParameter("id"));

        dao.deleteSubject(id);

        response.sendRedirect(
            request.getContextPath()
            + "/SubjectServlet?action=view");
    }

    // DEFAULT
    else {

        response.sendRedirect(
            request.getContextPath()
            + "/SubjectServlet?action=view");
    }
}

@Override
protected void doGet(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {

    processRequest(request, response);
}

@Override
protected void doPost(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {

    processRequest(request, response);
}
}

```

9.SubjectDAO.java
package com.lab.dao;

import com.lab.bean.SubjectBean;

```

import java.sql.*;
import java.util.ArrayList;

public class SubjectDAO {

    private Connection getConnection() throws Exception {

        Class.forName("com.mysql.jdbc.Driver");

        Connection con =
            DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/lab_7exercise",
                "root",
                "admin"
            );

        return con;
    }

    // ADD

    public void addSubject(SubjectBean subject) {

        try {

            Connection con = getConnection();

            String sql =
                "INSERT INTO registered_subjects "
                + "(matric_no, subject_code, subject_name) "
                + "VALUES (?, ?, ?)";

            PreparedStatement ps =
                con.prepareStatement(sql);

            ps.setString(1, subject.getMatricNo());
            ps.setString(2, subject.getSubjectCode());
            ps.setString(3, subject.getSubjectName());

            ps.executeUpdate();

            con.close();

        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    // VIEW

    public ArrayList<SubjectBean> getSubjects(String matricNo) {

```

```

ArrayList<SubjectBean> list =
    new ArrayList<>();

try {

    Connection con = getConnection();

    String sql =
        "SELECT * FROM registered_subjects "
        + "WHERE matric_no=?";

    PreparedStatement ps =
        con.prepareStatement(sql);

    ps.setString(1, matricNo);

    ResultSet rs = ps.executeQuery();

    while(rs.next()) {

        SubjectBean sb =
            new SubjectBean();

        sb.setId(rs.getInt("id"));
        sb.setMatricNo(rs.getString("matric_no"));
        sb.setSubjectCode(rs.getString("subject_code"));
        sb.setSubjectName(rs.getString("subject_name"));

        list.add(sb);
    }

    con.close();

} catch (Exception e) {
    e.printStackTrace();
}

return list;
}

// GET SUBJECT

public SubjectBean getSubjectById(int id) {

    SubjectBean sb =
        new SubjectBean();

    try {

        Connection con = getConnection();

```

```

String sql =
    "SELECT * FROM registered_subjects "
    + "WHERE id=?";

PreparedStatement ps =
    con.prepareStatement(sql);

ps.setInt(1, id);

ResultSet rs = ps.executeQuery();

if(rs.next()) {

    sb.setId(rs.getInt("id"));
    sb.setMatricNo(rs.getString("matric_no"));
    sb.setSubjectCode(rs.getString("subject_code"));
    sb.setSubjectName(rs.getString("subject_name"));
}

con.close();

} catch(Exception e) {
    e.printStackTrace();
}

return sb;
}

```

// UPDATE

```

public void updateSubject(SubjectBean subject) {

    try {

        Connection con = getConnection();

        String sql =
            "UPDATE registered_subjects "
            + "SET subject_code=?, subject_name=? "
            + "WHERE id=?";

        PreparedStatement ps =
            con.prepareStatement(sql);

        ps.setString(1, subject.getSubjectCode());
        ps.setString(2, subject.getSubjectName());
        ps.setInt(3, subject.getId());

        ps.executeUpdate();
    }
}

```

```
        con.close();

    } catch (Exception e) {
        e.printStackTrace();
    }
}

// DELETE

public void deleteSubject(int id) {

    try {

        Connection con = getConnection();

        String sql =
            "DELETE FROM registered_subjects "
            + "WHERE id=?";

        PreparedStatement ps =
            con.prepareStatement(sql);

        ps.setInt(1, id);

        ps.executeUpdate();

        con.close();

    } catch (Exception e) {
        e.printStackTrace();
    }
}
}
```

Output:

Student Dashboard

Welcome S001

[Register Subject](#)

[View Subjects](#)

1. Register Subject



The screenshot shows a web browser window with the address bar displaying 'localhost:8080/Lab7_Exercise/Subject/registerSubject.jsp'. The page title is 'Register Subject'. Below the title, there are two input fields: 'Subject Code:' with the value 'CSF3023' and 'Subject Name:' with the value 'Web'. Below these fields is a 'Register' button. At the bottom of the form, there is a link '[View Subjects](#)'.

2. Display Subject

← ↻ ⓘ localhost:8080/Lab7_Exercise/SubjectServlet?action=view

Subject List

[Add Subject](#)

ID	Code	Name	Action
13	CSF3023	Web	Edit Delete

3. Update Subject

← ↻ ⓘ localhost:8080/Lab7_Exercise/SubjectServlet?action=edit&id=13

Update Subject

Subject Code:

Subject Name:

4.Updated Subject

Subject List

[Add Subject](#)

ID	Code	Name	Action
13	CSF30231	Web Dev	Edit Delete

5. Delete Subject

✓

Subject List

[Add Subject](#)

ID	Code	Name	Action
----	------	------	--------